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(54) **CATHETER ACTUATOR WITH 360 DEGREE ACCESSIBILITY AND INTEGRATED LOCK**

(71) Applicant: **Boston Scientific Scimed, Inc.**, Maple Grove, MN (US)

(72) Inventors: **Dustin J. Olynyk**, Eagan, MN (US); **James Holmberg**, Champlin, MN (US); **Garth Mindermann, III**, Bloomington, MN (US); **Michael Strawn**, Minneapolis, MN (US)

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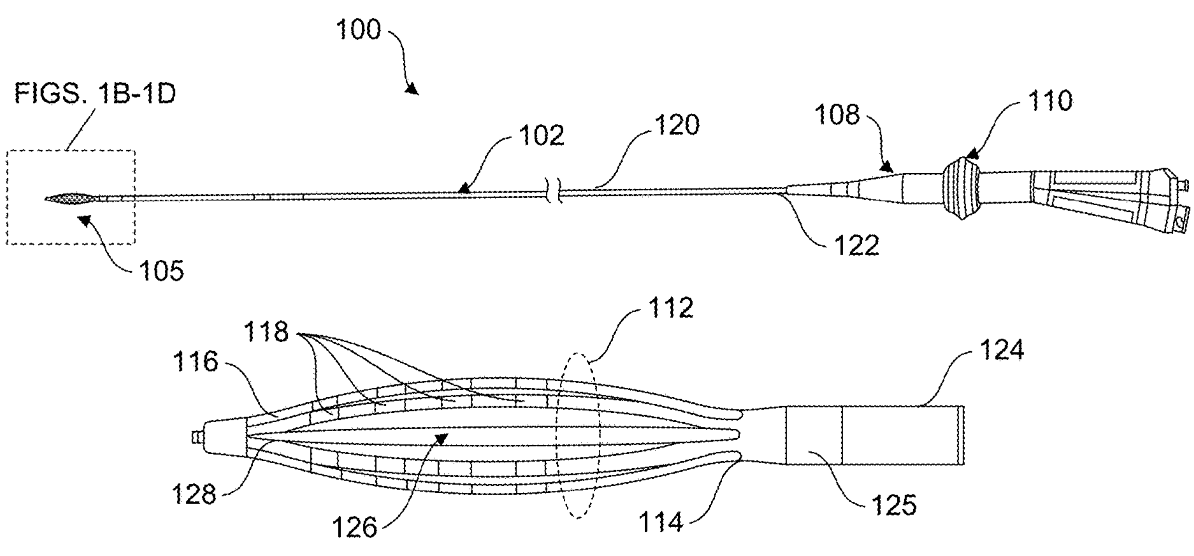
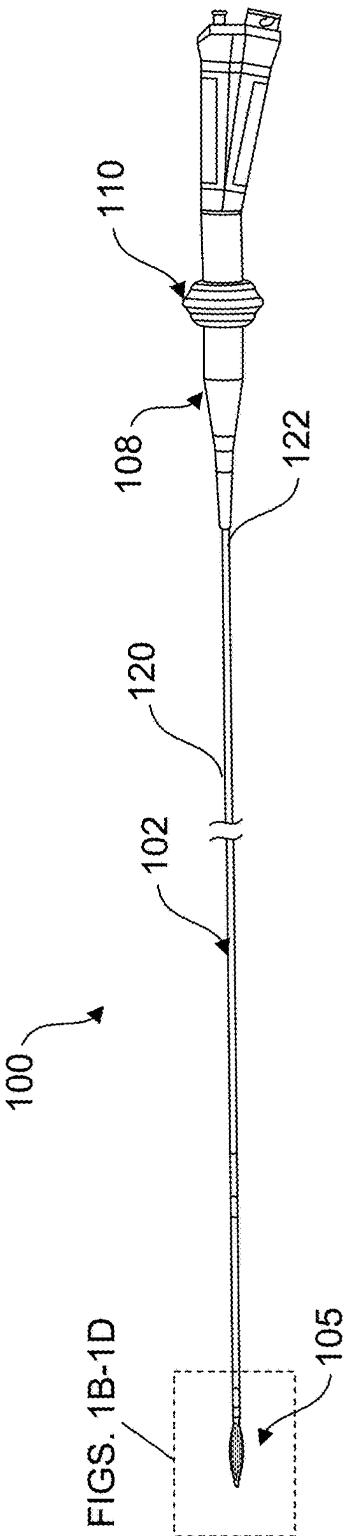
(57) **ABSTRACT**

A medical catheter comprising a tubular outer shaft, an inner shaft slidably disposed within the outer shaft and dimensioned to extend distally beyond the outer shaft, a radially expandable and radially collapsible spline assembly comprising a plurality of splines, a handle coupled to the proximal end of the outer shaft, wherein the proximal portion of the inner shaft is slidably disposed within the handle, an actuator comprising a slider knob slidably disposed about and fully circumscribing the outer surface of the handle body, wherein the slider knob is operatively coupled to the proximal portion of the inner shaft such that axial displacement of the slider knob by the user causes the inner shaft to translate axially relative to the outer shaft to selectively control a configuration of the spline assembly.

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FIGS. 1B-1D

FIG. 1A

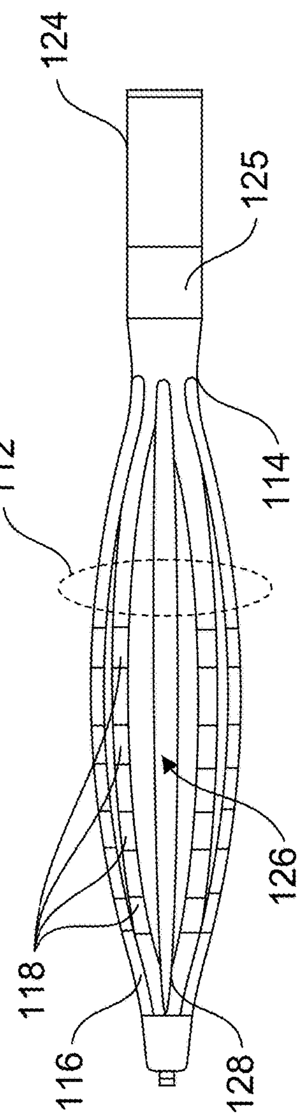
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FIG. 1B

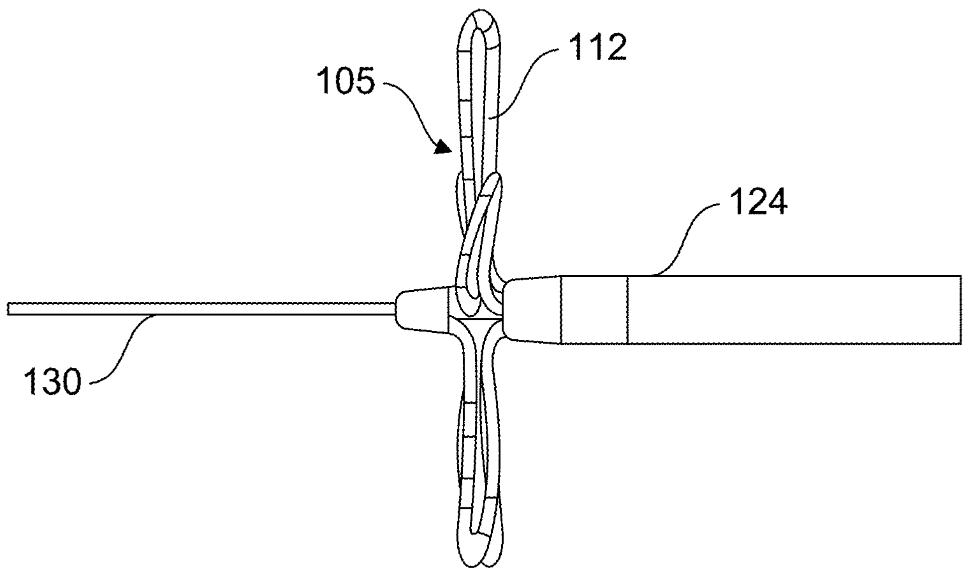

FIG. 1C

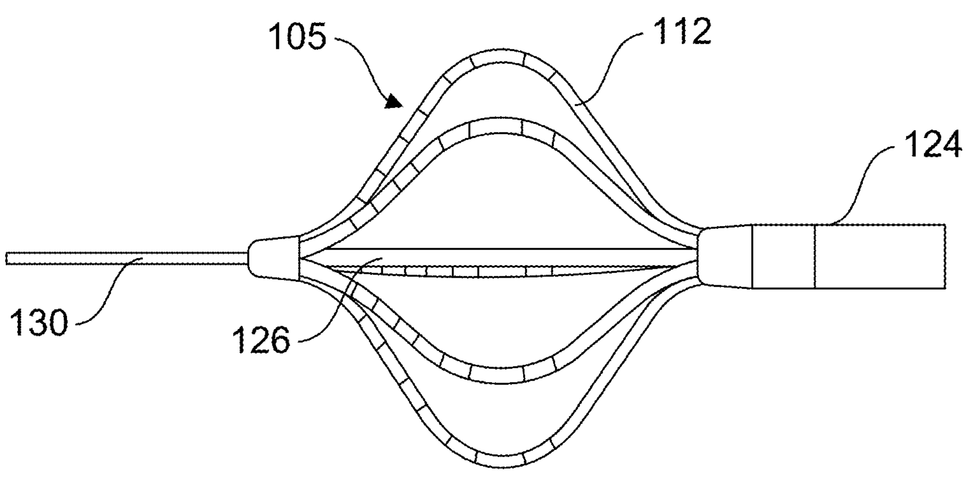

FIG. 1D

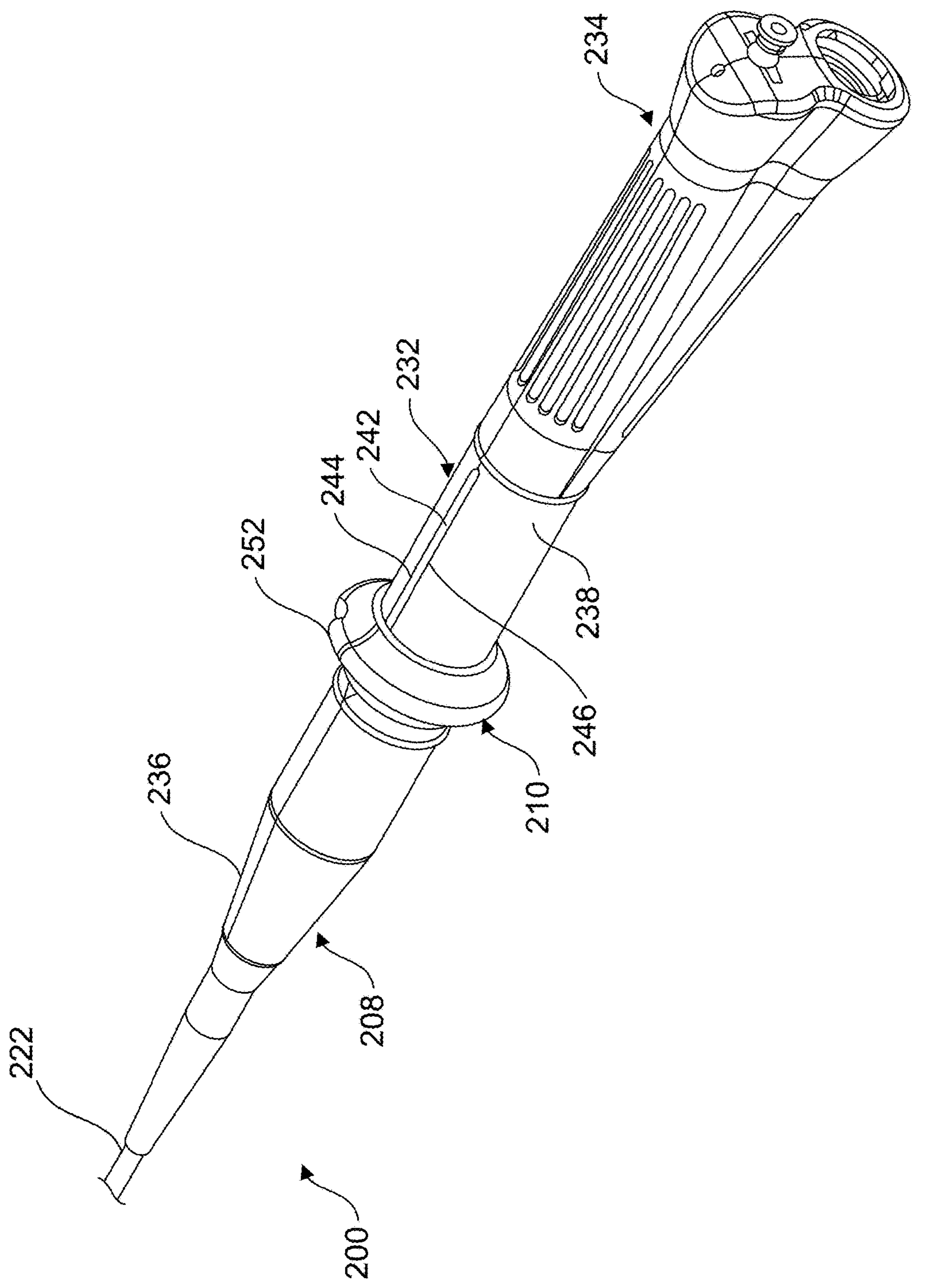

FIG. 2A

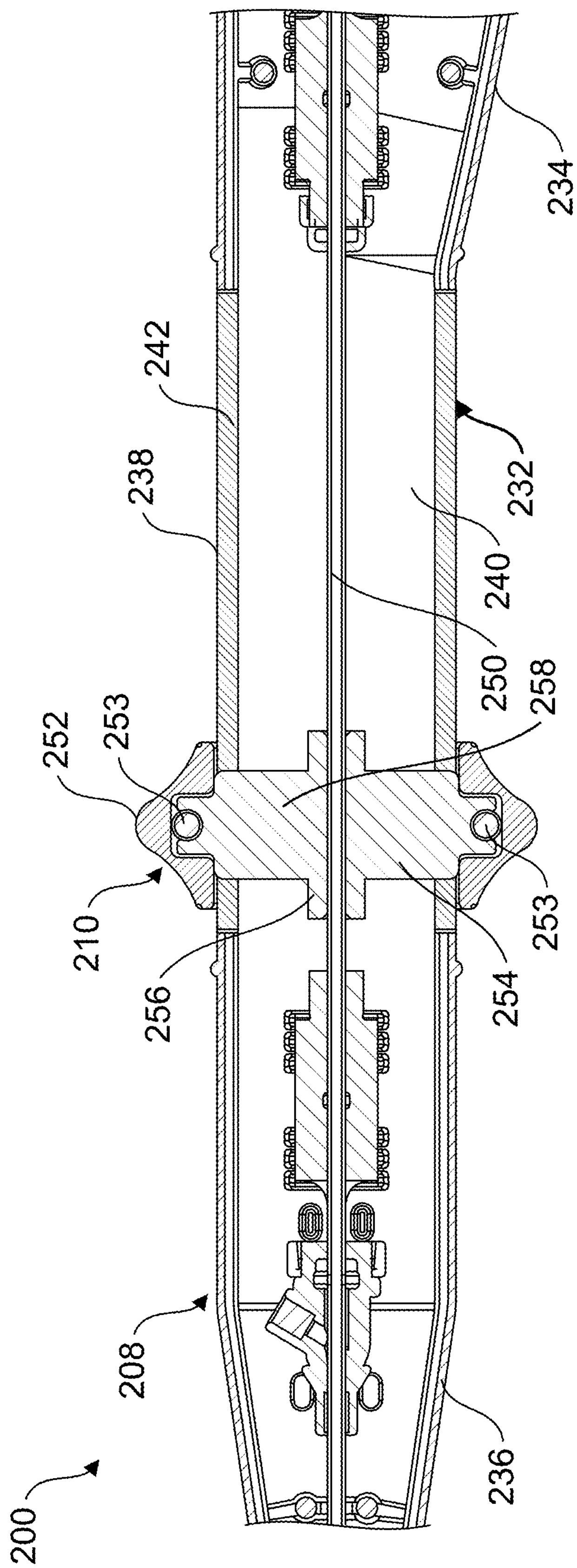

FIG. 2B

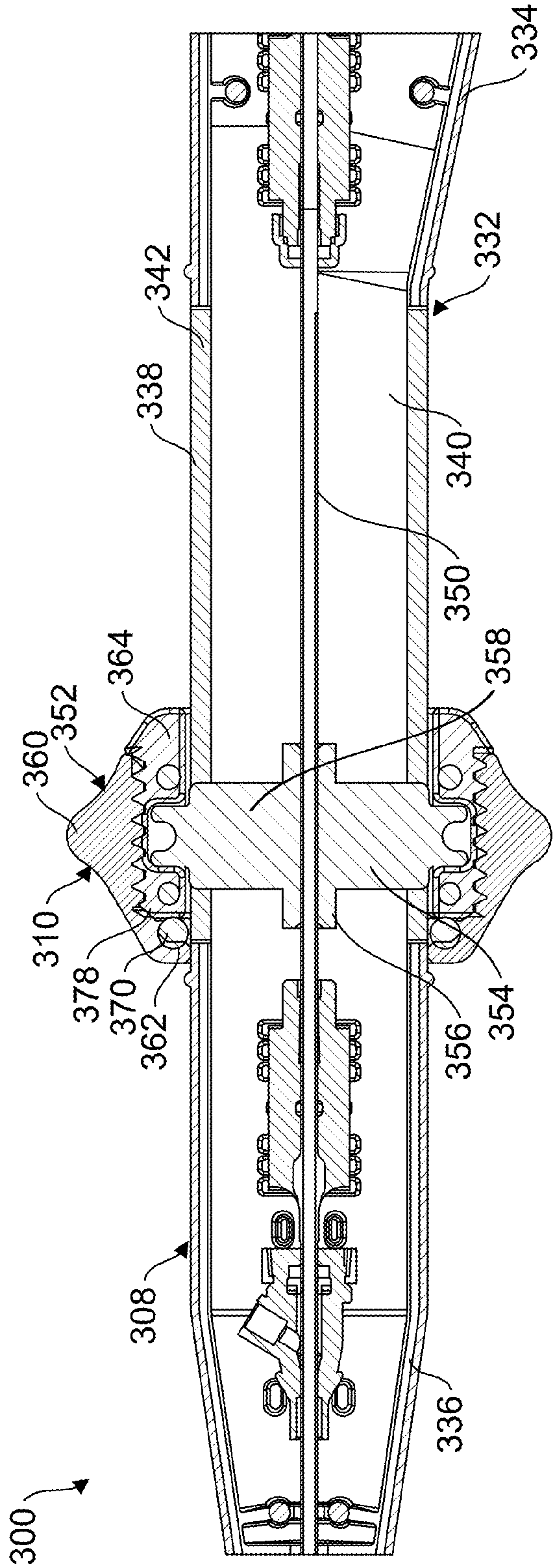

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FIG. 3

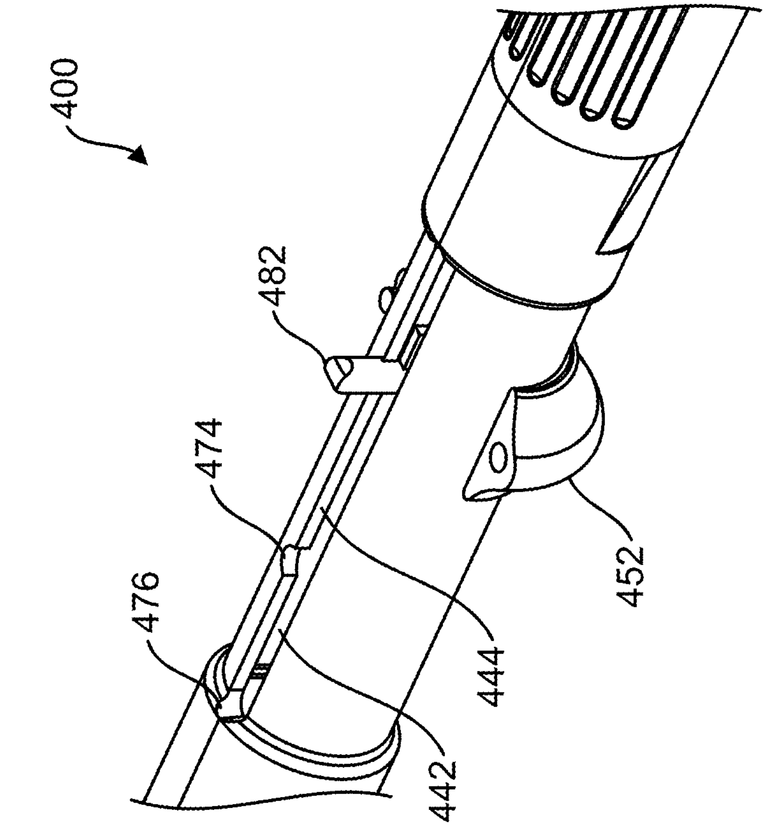

FIG. 4B

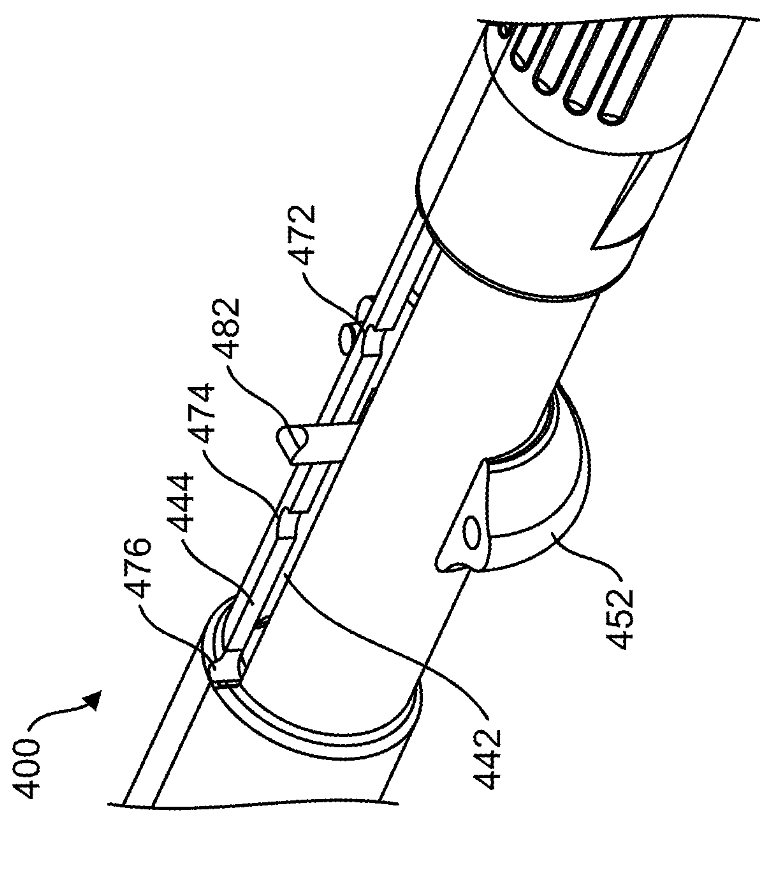

FIG. 4A

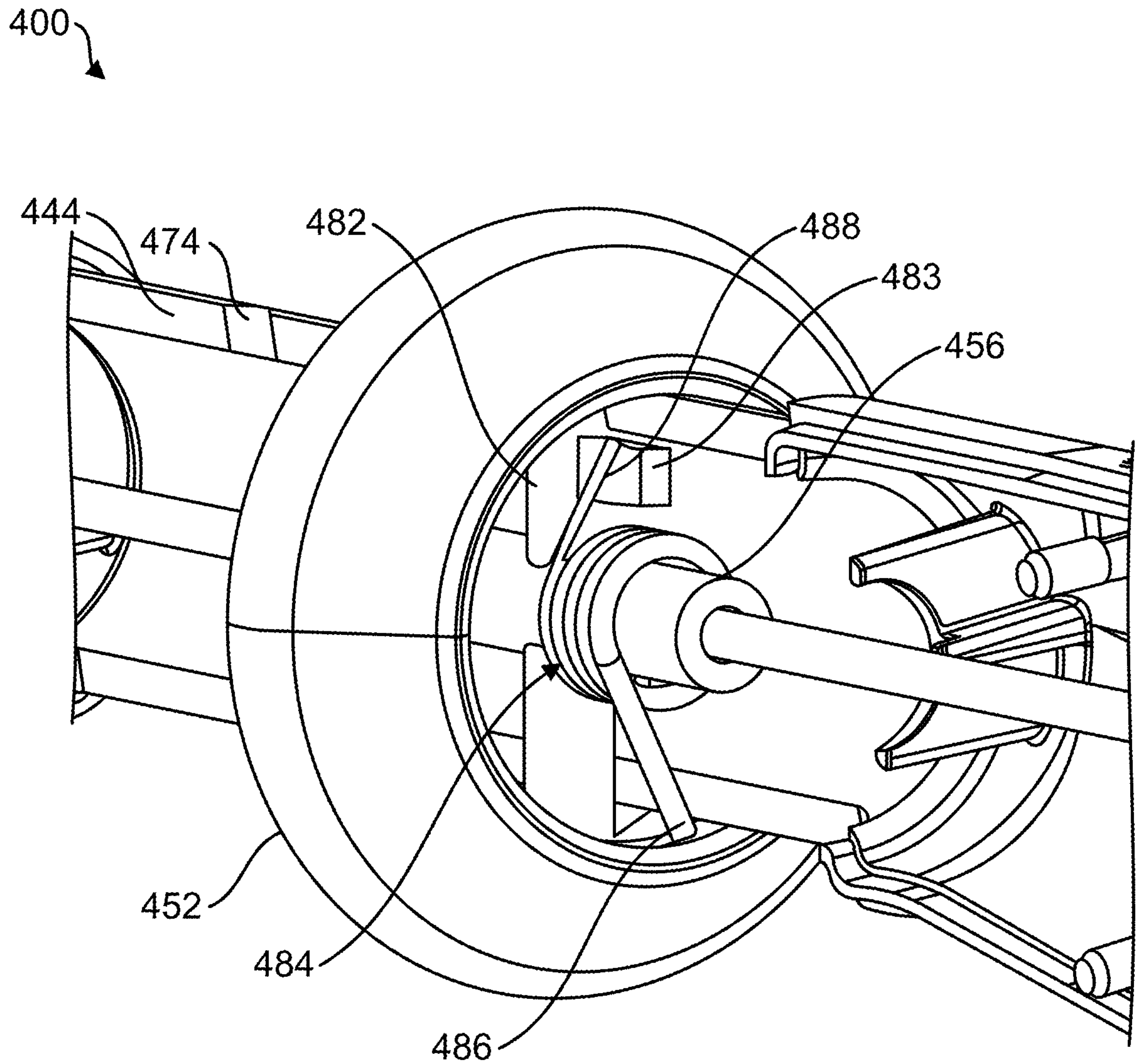

FIG. 4C

CATHETER ACTUATOR WITH 360 DEGREE ACCESSIBILITY AND INTEGRATED LOCK

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to U.S. Provisional Patent Application No. 63/563,089, filed Mar. 8, 2024, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to medical systems and methods for cardiac ablation, and more particularly, to ablation catheters for use in ablating cardiac tissues.

BACKGROUND

[0003] Ablation procedures (e.g., pulsed field ablation, radiofrequency ablation, cryoablation, and the like) are used to treat many different conditions in patients. Ablation can be used to treat cardiac arrhythmias, benign tumors, cancerous tumors, and to control bleeding during surgery. There is a continuing need for ablation catheters with features that improve usability and clinical efficiency.

SUMMARY

[0004] In Example 1, a medical catheter comprising a tubular outer shaft, an inner shaft, a radially expandable and radially collapsible spline assembly, a handle and an actuator. inner shaft is slidably disposed within the outer shaft and is dimensioned to extend distally beyond the outer shaft. The spline assembly comprises a plurality of splines each having a spline proximal end attached to the distal end of the outer shaft, and an opposite spline distal end attached to the distal end of the inner shaft. The handle has a handle proximal end, and a handle distal end, wherein the proximal end of the outer shaft is attached to and extends from the handle proximal end, and a proximal portion of the inner shaft is slidably disposed within the handle. The actuator comprises a slider knob slidably disposed about and fully circumscribing the outer surface of the handle, wherein the slider knob is operatively coupled to the proximal portion of the inner shaft such that axial displacement of the slider knob by a user causes the inner shaft to translate axially relative to the outer shaft to selectively control a configuration of the spline assembly.

[0005] In Example 2, the medical catheter of Example 1, wherein the actuator further comprises an inner actuator member attached to and extending from the slider knob into the handle, and wherein the inner actuator member is fixedly secured to the proximal portion of the inner shaft within the handle and operatively couples the inner shaft to the slider knob.

[0006] In Example 3, the medical catheter of Example 2, wherein the inner actuator member includes a hub disposed about and fixedly secured to the proximal portion of the inner shaft, and a radial extension extending from the hub to the slider knob.

[0007] In Example 4, the medical catheter of Example 3, wherein the handle includes an axial slot defined by opposed first and second lateral edges, and wherein the radial extension extends through the axial slot.

[0008] In Example 5, the medical catheter of Example 4, wherein the first and second lateral edges constrain rotation of the inner actuator member during axial displacement of the slider knob.

[0009] In Example 6, the medical catheter of any of Examples 1-5, wherein the actuator is configured such that the user can lock the slider knob in an axial position.

[0010] In Example 7, the medical catheter of any of Examples 1-6, wherein the slider knob includes an outer knob element that is rotatable about the handle, wherein rotation of the outer knob element by the user in a first direction locks the axial position of the slider knob, and rotation of the outer knob element in a second direction opposite the first direction permits the slider knob to be axially displaced.

[0011] In Example 8, the medical catheter of Example 7, wherein the slider knob further comprises an inner knob element that is rotationally fixed relative to the handle and includes a forward portion disposed radially inward of and in threaded engagement with the outer knob element, wherein rotation of the outer knob element causes the inner knob element to translate axially to selectively lock and unlock the slider knob relative to the handle.

[0012] In Example 9, the medical catheter of Example 8, wherein the outer knob element includes a forward inner surface, and wherein the handle includes a compression element disposed about the handle between the forward inner surface and the forward portion of the inner knob element, wherein rotation of the outer knob element in the first direction causes the inner knob element to move in a distal direction to compress the compression element between the handle, the forward inner surface of the outer knob element, and the forward portion of the inner knob element to lock the axial position of the slider knob.

[0013] In Example 10, the medical catheter of Example 9, wherein the compression element is an O-ring.

[0014] In Example 11, the medical catheter of any of Examples 1-5, wherein the actuator further comprises a locking arrangement configured to releasably lock the actuator in one or more pre-defined axial positions.

[0015] In Example 12, the medical catheter of Example 11, wherein the first lateral edge includes a first detent at a first location along the lateral edge, and wherein the locking arrangement comprises a locking pin extending radially from the hub through the axial slot, wherein the locking pin is biased toward the first lateral edge so as to automatically engage the first detent to releasably lock the axial position of the slider knob corresponding to the first location.

[0016] In Example 13, the medical catheter of Example 12, wherein the locking arrangement further includes a torsion spring disposed about the hub, the torsion spring including a first free end frictionally engaged with an inner surface of the handle, and an opposite second free end positioned to engage and apply a biasing force to the locking pin to bias the locking pin toward the first lateral edge.

[0017] In Example 14, the medical catheter of Example 13, wherein the slider knob is configured to be rotatable to counteract the biasing force and selectively disengage the locking pin from the first detent.

[0018] In Example 15, the medical catheter of Example 14, wherein the first lateral edge includes a second detent at a second location along the first lateral edge, and a third detent at a third location along the first lateral edge, the first, second and third locations being spaced from one another and corresponding to pre-determined configurations of the spline assembly.

[0019] In Example 16, a medical catheter comprising a catheter body, a spline assembly, a handle and an actuator. The catheter body comprises a tubular outer shaft having a proximal end and an open distal end opposite the proximal end, and an inner shaft slidably disposed within the outer shaft having a proximal portion and an opposite distal end, the inner shaft being dimensioned such that the inner shaft distal end extends distally beyond the distal end of the outer shaft. The spline assembly comprises a plurality of splines each having a spline proximal end attached to the distal end of the outer shaft, and an opposite spline distal end attached to the distal end of the inner shaft, the spline assembly configured to be transitioned between a collapsed configuration dimensioned for slidable insertion through a delivery sheath, and an expanded configuration. The handle comprises a handle body having a body proximal end, a body distal end, an outer surface extending from the body proximal end to the body distal end, the handle body defining an interior chamber, wherein the proximal end of the outer shaft is attached to and extends from the body proximal end, and the proximal portion of the inner shaft is disposed within the interior chamber. The actuator is slidably coupled to the handle body between the body proximal end and the body distal end, the actuator comprising a slider knob disposed about and fully circumscribing the outer surface of the handle body such that the slider knob can be axially displaced by a user, wherein the slider knob is further operatively coupled to the proximal portion of the inner shaft such that axial displacement of the slider knob by the user causes the inner shaft to translate axially relative to the outer shaft to transition the spline assembly between the collapsed configuration and the expanded configuration.

[0020] In Example 17, the medical catheter of Example 16, wherein the actuator further comprises an inner actuator member attached to and extending from the slider knob into the interior chamber of the handle, and wherein the inner actuator member is fixedly secured to the proximal portion of the inner shaft within the interior chamber and operatively couples the inner shaft to the slider knob.

[0021] In Example 18, the medical catheter of Example 17, wherein the inner actuator member includes a hub disposed about and fixedly secured to the proximal portion of the inner shaft, and a radial extension extending from the hub to the slider knob.

[0022] In Example 19, the medical catheter of Example 18, wherein the handle body includes an axial slot defined by opposed first and second lateral edges, and wherein the radial extension extends through the axial slot and the first and second lateral edges constrain rotation of the inner actuator member during axial displacement of the slider knob.

[0023] In Example 20, the medical catheter of Example 19, wherein the actuator is configured such that the user can lock the slider knob in an axial position.

[0024] In Example 21, the medical catheter of Example 20, wherein the slider knob includes an outer knob element that is rotatable about the handle body, wherein rotation of the outer knob element by the user in a first direction locks the axial position of the slider knob, and rotation of the outer knob element in a second direction opposite the first direction permits the slider knob to be axially displaced.

[0025] In Example 22, the medical catheter of Example 21, wherein the slider knob further comprises an inner knob element that is rotationally fixed relative to the handle body and includes a forward portion disposed radially inward of and in threaded engagement with the outer knob element, wherein rotation of the outer knob element causes the inner knob element to translate axially to selectively lock and unlock the slider knob relative to the handle body.

[0026] In Example 23, the medical catheter of Example 22, wherein the outer knob element includes a forward inner surface, and wherein the handle includes a compression element disposed about the handle body between the forward inner surface and the forward portion of the inner knob element, wherein rotation of the outer knob element in the first direction causes the inner knob element to move in a distal direction to compress the compression element between the handle body outer surface, the forward inner surface of the outer knob element, and the forward portion of the inner knob element to lock the axial position of the slider knob.

[0027] In Example 24, the medical catheter of Example 19, wherein the first lateral edge includes a first detent at a first location along the lateral edge, and wherein the locking arrangement comprises a locking pin extending radially from the hub through the axial slot, wherein the locking pin is biased toward the first lateral edge so as to automatically engage the first detent to releasably lock the axial position of the slider knob corresponding to the first location.

[0028] In Example 25, the medical catheter of Example 24, wherein the locking arrangement further includes a torsion spring disposed about the hub, the torsion spring including a first free end frictionally engaged with an inner surface of the handle body, and an opposite second free end positioned to engage and apply a biasing force to the locking pin to bias the locking pin toward the first lateral edge.

[0029] In Example 26, the medical catheter of Example 25, wherein the slider knob is configured to be rotatable to counteract the biasing force and selectively disengage the locking pin from the first detent.

[0030] In Example 27, the medical catheter of Example 26, wherein the first lateral edge includes a second detent at a second location along the first lateral edge, and a third detent at a third location along the first lateral edge, the first, second and third locations being spaced from one another and corresponding to pre-determined configurations of the spline assembly.

[0031] In Example 28, a medical catheter comprising a catheter body, a radially expandable and radially collapsible spline assembly, a handle and an actuator. The catheter body comprises a tubular outer shaft and an inner shaft slidably disposed within the outer shaft and dimensioned to extend distally beyond the outer shaft. The spline assembly comprises a plurality of splines each having a spline proximal end attached to the distal end of the outer shaft, and an opposite spline distal end attached to the distal end of the inner shaft. The handle is coupled to the catheter body, the handle having a handle proximal end and a handle distal end, wherein a proximal end of the outer shaft is attached to and extends from the handle proximal end, and a proximal portion of the inner shaft is slidably disposed within the handle. The actuator comprises a slider knob slidably disposed about and fully circumscribing an outer surface of the handle, wherein the slider knob is operatively coupled to the proximal portion of the inner shaft such that axial displacement of the slider knob by a user causes the inner shaft to translate axially relative to the outer shaft to selectively control an amount of expansion of the spline assembly.

[0032] In Example 29, the medical catheter of Example 28, wherein the actuator is configured such that the user can lock the slider knob in an axial position.

[0033] In Example 30, the medical catheter of Example 28, wherein the actuator further comprises a locking arrangement configured to releasably lock the actuator in one or more pre-defined axial positions.

[0034] In Example 31, a handle assembly for a medical catheter having a catheter body including a tubular outer shaft, an inner shaft slidably disposed within the outer shaft, and a radially expandable and collapsible splined electrode assembly disposed at a distal end of the catheter body. The handle assembly comprises a handle and an actuator. The handle comprises a handle body having a body proximal end, a body distal end, an outer surface extending from the body proximal end to the body distal end, the handle body defining an interior chamber configured to slidably receive a proximal portion of the inner shaft. The actuator is slidably coupled to the handle body between the body proximal end and the body distal end, the actuator comprising a slider knob disposed about and fully circumscribing the outer surface of the handle body such that the slider knob can be axially displaced by a user, wherein the slider knob is further configured to be operatively coupled to the proximal portion of the inner shaft such that axial displacement of the slider knob by the user can cause the inner shaft to translate axially relative to the outer shaft to transition the spline assembly between a collapsed configuration and an expanded configuration.

[0035] In Example 32, the handle assembly of Example 31, wherein the handle body includes an axial slot defined by opposed first and second lateral edges, and the actuator further comprises an inner actuator member including a hub configured to be disposed about and fixedly secured to the proximal portion of the inner shaft, and a radial extension extending from the hub through the axial slot to the slider knob.

[0036] In Example 33, the handle assembly of Example 32, wherein the slider knob includes an outer knob element that is rotatable about the handle body, an inner knob element that is rotationally fixed relative to the handle body and includes a forward portion disposed radially inward of and in threaded engagement with the outer knob element, wherein rotation of the outer knob element causes the inner knob element to translate axially to selectively lock and unlock the slider knob relative to the handle body.

[0037] In Example 34, the handle assembly of Example 33, wherein the outer knob element includes a forward inner surface, and wherein the handle further includes a compression element disposed about the handle body between the forward inner surface and the forward portion of the inner knob element, wherein rotation of the outer knob element in a first direction causes the inner knob element to move in a distal direction to compress the compression element between the handle body outer surface, the forward inner surface of the outer knob element, and the forward portion of the inner knob element to lock the axial position of the slider knob.

[0038] In Example 35, the handle assembly of Example 32, wherein the first lateral edge includes a plurality of axially-spaced detents, and wherein the actuator includes a locking arrangement comprises a locking pin extending radially from the hub through the axial slot, wherein the locking pin is biased toward the first lateral edge so as to automatically engage the one of the plurality of detents to releasably lock the axial position of the slider knob corresponding to a predetermined configuration of the spline assembly.

[0039] While multiple embodiments are disclosed, still other embodiments of the present disclosure will become apparent to those skilled in the art from the following detailed description, which shows and describes illustrative embodiments of the disclosure. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

[0040] FIG. **1**A is a plan view of an exemplary ablation catheter having a splined ablation electrode assembly according to embodiments of the present disclosure.

[0041] FIG. **1**B is a plan view of the distal end portion of the ablation catheter of FIG. **1**A, showing details of the splined ablation electrode assembly, according to embodiments of the present disclosure.

[0042] FIGS. **1**C-**1**D are plan views of the distal end portion of the ablation catheter of FIG. **1**A showing the spline ablation electrode assembly in various states of deployment.

[0043] FIGS. **2**A and **2**B are partial isometric and cross-sectional views, respectively, of a handle and actuator assembly for an ablation catheter, such as the ablation catheter of FIG. **1**A, according to embodiments of the present disclosure.

[0044] FIG. **3** is a partial cross-sectional view of an alternative handle and actuator assembly for an ablation catheter, such as the ablation catheter of FIG. **1**A, according to embodiments of the present disclosure.

[0045] FIGS. **4**A-**4**C are partial isometric views of an alternative handle and actuator assembly for an ablation catheter, such as the ablation catheter of FIG. **1**A, according to embodiments of the present disclosure.

[0046] While the disclosure is amenable to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and are described in detail below. The intention, however, is not to limit the disclosure to the particular embodiments described. On the contrary, the disclosure is intended to cover all modifications, equivalents, and alternatives falling within the scope of the disclosure as defined by the appended claims.

DETAILED DESCRIPTION

[0047] For purposes of promoting an understanding of the principles of the present disclosure, reference is now made to the examples illustrated in the drawings, which are described below. The illustrated examples disclosed herein are not intended to be exhaustive or to limit the disclosure to the precise form disclosed in the following detailed description. Rather, these exemplary embodiments were chosen and described so that others skilled in the art may use their teachings. It is not beyond the scope of this disclosure to have a number (e.g., all) the features in a given example used across all examples. Thus, no one figure should be interpreted as having any dependency or requirement related to any single component or combination of components illustrated therein. Additionally, various components depicted in a given figure may be, in examples, integrated with various ones of the other components depicted therein (and/or components not illustrated), all of which are considered to be within the ambit of the present disclosure.

[0048] FIG. **1**A is a plan view of an exemplary ablation catheter **100** having a splined assembly **105** according to embodiments of the present disclosure. FIG. **1**B is a plan view of the distal end portion of the ablation catheter **100**, showing, among other things, details of the spline assembly **105**, according to embodiments of the present disclosure. In various embodiments, the ablation catheter may be particularly useful in ablation procedures targeting the pulmonary vein ostium of a patient for treating atrial fibrillation. In particular, the ablation catheter **100** may be configured as a pulsed field ablation catheter with electrodes configured to receive pulsed electrical signals/waveforms from a pulse generator, thereby creating pulsed electric fields sufficient for ablating target tissue via irreversible electroporation.

[0049] Referring collectively to FIGS. **1**A and **1**B, as shown, the ablation catheter **100** includes a catheter body **102**, the spline assembly **105**, a handle **108** and an actuator **110**. As further shown, the catheter body **102** extends from the handle **108**, and the spline assembly **105** is attached to and extends distally from the catheter body **102**. Additionally, the actuator **110** is coupled to the handle **108**. As will be explained in greater detail herein, the actuator **110** is movable (i.e., slidable) relative to the handle **108** under action by a clinician and is operable to change the geometric configuration of the spline assembly **105** during use. Additionally, the configuration of the various embodiments of the actuator **110** of the present disclosure facilitate improved ease of use by the clinician, in that the actuator **110** is readily accessible by the user regardless of the orientation of the handle **108**.

[0050] With particular reference to FIG. **1**B, in the illustrated embodiment, the spline assembly **105** includes a plurality of splines **112** each having a spline proximal end **114** and an opposite spline distal end **116**, with each spline carrying a plurality of electrodes **118**. For clarity of illustration, only a single spline **112** is labeled in FIG. **1**B, although the skilled artisan will readily recognize that the referenced features apply to all of the illustrated splines. The skilled artisan will also readily recognize that the particular number of splines **112** and electrodes **118** illustrated in FIG. **1**B is merely exemplary, and thus embodiments with more or fewer splines **112** or electrodes **118** fall within the scope of the present disclosure. Additionally, the size and shape of the illustrated electrodes **118** is similarly exemplary. In embodiments, the electrodes **118** may be operable as ablation electrodes, sensing electrodes, or tracking electrodes, e.g., as used for impedance tracking of the location of the spline assembly **105** in electroanatomical mapping systems, or combinations of the foregoing.

[0051] As further shown, the catheter body **102** includes an outer shaft **120** having a proximal end **122**, a distal end **124** opposite the proximal end **122**, and optionally, a shaft electrode **125** located proximate the distal end **124**. Additionally, the catheter body **102** includes an inner shaft **126** having a distal end **128**. As shown, the proximal end **122** of the outer shaft **120** is attached to and extends distally from the handle **108**.

[0052] In embodiments, the outer shaft **120** is tubular so as to define an outer shaft lumen (not shown) extending from the proximal end **122** through the distal end **124** such that the distal end **124** is open. Additionally, the inner shaft **126** is slidably disposed within the outer shaft and is dimensioned such that the inner shaft distal end **128** extends distally beyond the distal end **124** of the outer shaft **120**. As further shown, the proximal end **114** of each spline **112** is attached to the distal end **124** of the outer shaft **120**, and the distal end **116** of each spline **112** is attached to the distal end **128** of the inner shaft **126**. Accordingly, the spline assembly **105** is configured to be transitioned between a radially collapsed configuration, such as shown in FIG. **1**B, whereby the spline assembly **105** is dimensioned for slidable insertion through a delivery sheath (not shown), and a radially expanded configuration, as illustrated and explained further elsewhere. Additionally, as will be explained in greater detail elsewhere herein, the inner shaft **126** is coupled to the actuator **110**, such that the user can change the relative axial position of the distal ends **128**, **124** of the inner shaft **126** and the outer shaft **120**, respectively, via manipulation of the actuator **110**. In this way, the aforementioned transition of the spline assembly **105** configuration is accomplished through axial extension and retraction of the inner shaft **124** relative to the outer shaft **120** using the actuator **110**.

[0053] FIGS. **1**C-**1**D are plan views of the distal end portion of the ablation catheter **100** showing the spline assembly **105** in different exemplary states of deployment. In the configuration illustrated in FIG. **1**C, the distal end **128** of the inner shaft **126** is fully retracted relative to the distal end **124** of the outer shaft **120**, such that the spline assembly **105** assumes a minimum axial length. In this configuration, the splines **112** each form a petal-shaped configuration. In embodiments, the spline assembly **105** can assume a form factor similar to that illustrated in FIGS. **26**A-**26**E, **28**A-**28**E and **29**A-**29**B, and described in the corresponding detailed description, of commonly-assigned U.S. Pat. No. 10,172,673, which is incorporated herein by reference in its entirety for all purposes. The fully expanded configuration illustrated in FIG. **1**C is particularly suited for spanning a target pulmonary vein from within the left atrium for performing a pulmonary vein isolation via pulsed field ablation.

[0054] FIG. **1**D illustrates the spline assembly **105** in a partially expanded configuration. In this configuration, the axial position of the distal end **128** of the inner shaft **126** is intermediate that of the fully extended position shown in FIG. **1**B, and the fully retracted position shown in FIG. **1**C. The geometry of the spline assembly **105** illustrated in FIG. **1**D may, in some respects, be similar to conventional splined basket catheter arrangements as are known in the relevant arts.

[0055] Retraction and extension of the distal end **128** of the inner shaft **126** relative to the outer shaft **120** is controlled by the user through axial displacement (either proximally or distally) of the actuator **110**.

[0056] In the illustrated embodiment, the inner shaft **126** includes a lumen so as to be configured to receive and be deployed over a guide wire **130**, although in embodiments this feature is omitted.

[0057] FIGS. **2**A and **2**B are partial isometric and cross-sectional views, respectively, of an ablation catheter **200** according to embodiments of the present disclosure. The ablation catheter **200** is identical to the catheter **100** and includes a handle **208** and an actuator assembly **210** for changing the configuration, e.g., geometry, of an expandable spline assembly (e.g., the spline assembly **105** of the catheter **100**).

[0058] Referring to FIGS. **2**A and **2**B collectively, as shown, the handle **208** includes a handle body **232** having a body proximal end **234**, a body distal end **236**, an outer surface **238** extending from the body proximal end **234** to the body distal end **236**. The handle body **232** is in the form of a shell defining an interior chamber **242**. As further shown, the handle body **232** includes an axial slot **242** bounded by opposed lateral edges **244**, **246**. Additionally, a proximal portion **250** of the inner shaft extends through the handle body **232** within the interior chamber **242**. As further shown, a proximal end **222** of the outer shaft is attached to and extends from the body proximal end **234**.

[0059] The skilled artisan will recognize that FIGS. **2**A and **2**B illustrate additional features, e.g., flushing components, electrical and electronic components, and the like that are not critical to the present disclosure and are thus not described in detail.

[0060] In embodiments, the actuator **210** is slidably coupled to the handle body **232** between the body proximal end **234** and the body distal end **236**. As shown, the actuator **210** includes a slider knob **252** disposed about and fully circumscribing the outer surface **238** of the handle body **232**. In the various embodiments, the slider knob **252** is configured to be axially displaceable by a user, i.e., in the proximal-distal direction of the handle body **232**.

[0061] In the illustrated embodiment, the slider knob **252** comprises two pieces, e.g., half-shells, to facilitate assembly, which are secured together by fasteners **253**, e.g., screws. In other embodiments, the portions of the slider knob **252** may be secured together by other means, e.g., snap-fit connections, adhesive, and the like.

[0062] As further shown, the actuator **210** additionally includes an inner actuator member **254** attached to and extending from the slider knob **252** into the interior chamber **240** of the handle **208**. In the illustrated embodiment, the inner actuator member **254** includes a hub **256** and a radial extension **258**. As shown, the hub **256** is disposed about and fixedly secured to the proximal portion **250** of the inner shaft, and the radial extension **258** extends from the hub **256** through the axial slot **242** to the slider knob **252**. As such, the slider knob **252** is operatively coupled to the proximal portion **250** of the inner shaft such that axial displacement of the slider knob **252** by the user causes the inner shaft to translate axially relative to the outer shaft **222**, which in turn changes the configuration of the spline assembly as described in connection with FIGS. **1**B-**1**D. In embodiments, the axial slot **242** has a width that is sized to allow axial movement of the slider knob **252** while constraining rotation of the inner actuator member **254** during axial displacement of the slider knob **252**.

[0063] Typical clinical procedures utilizing an ablation catheter such as the catheter **200** require the user to manipulate, e.g., rotate the handle **208** to place the spline assembly and corresponding electrodes at the desired location and orientation relative to the target tissue site. In existing ablation catheters having spline assemblies similar to those of the present disclosure, the mechanism for displacing the inner shaft is typically limited to a slider member extending from only a portion of the circumference of the handle. As such, in some instances the aforementioned slider member may not consistently be accessible to the user without difficulty. In contrast, as can be seen particularly in FIG. **2**A, the configuration of the actuator **210** of the ablation catheter **200**, with the slider knob **252** fully encircling or circumscribing the outer surface **238** of the handle body **232**, greatly enhances usability of the ablation catheter **200** by providing the user with free access to the slider knob **252** regardless of the overall orientation of the handle **208**.

[0064] FIG. **3** is a partial cross-sectional view of an ablation catheter **300** according to embodiments of the present disclosure. The ablation catheter **300** is identical to the catheters **100** and **200** except as described and illustrated in connection with FIG. **3**. In particular, the ablation catheter **300** has an actuation means for controlling the configuration of the spline assembly that includes an integrated locking arrangement for releasably and selectively locking the axial position of the actuator, and consequently, the configuration of the spline assembly.

[0065] As shown, the ablation catheter **300** includes a handle **308** and an actuator assembly **310** for changing the configuration, e.g., geometry, of an expandable spline assembly (e.g., the spline assembly **105** of the catheter **100**). The handle **308** includes a handle body **332** having a body proximal end **334**, a body distal end **336**, an outer surface **338** extending from the body proximal end **334** to the body distal end **336**. The handle body **332** is in the form of a shell defining an interior chamber **342**. As further shown, the handle body **332** includes an axial slot **342**. Additionally, a proximal portion **350** of the inner shaft extends through the handle body **332** within the interior chamber **342**.

[0066] In embodiments, the actuator **310** is slidably coupled to the handle body **332** between the body proximal end **334** and the body distal end **336**. As shown, the actuator **310** includes a slider knob **352** disposed about and fully circumscribing the outer surface **338** of the handle body **332** and configured to be axially displaceable by a user, i.e., in the proximal-distal direction of the handle body **332**.

[0067] As further shown, the actuator **310** additionally includes an inner actuator member **354** attached to and extending from the slider knob **352** into the interior chamber **340** of the handle **308**. In the illustrated embodiment, the inner actuator member **354** includes a hub **356** and a radial extension **358**. As shown, the hub **356** is disposed about and fixedly secured to the proximal portion **350** of the inner shaft, and the radial extension **358** extends from the hub **356** through the axial slot **342** to the slider knob **352**. As such, the slider knob **352** is operatively coupled to the proximal portion **350** of the inner shaft such that axial displacement of the slider knob **352** by the user causes the inner shaft to translate axially relative to the outer shaft (not shown), which in turn changes the configuration of the spline assembly as described in connection with FIGS. **1**B-**1**D.

[0068] As shown in FIG. **3**, the slider knob **352** includes an outer knob element **360** having a forward inner surface **362**, and an inner knob element **364** having a forward portion **368** in threaded engagement with the outer knob element **360**. The outer knob element **360** is rotatable about the handle body **332**. The inner knob element **364** is securely attached to the radial extension **358** of the inner actuator member **354** and thus is rotationally fixed relative to the handle body **332**. As further shown, a compression element **370**, e.g., an O-ring or gasket, is disposed about the handle body **332** and captured between the forward inner surface **362** of the outer knob element **362** and the forward portion **368** of the inner knob element **364**.

[0069] Accordingly, because the inner knob element **364** cannot rotate relative to the handle body **332**, rotation of the outer knob element **360** in a first direction causes the inner knob element **364** to move in a distal direction (due to the threaded engagement between the outer and inner knob elements **360**, **364**) to compress the compression element **370** between the handle body outer surface **338**, the forward inner surface **362** of the outer knob element **360**, and the forward portion **368** of the inner knob element **364** to lock the axial position of the slider knob **352**. Additionally, rotation of the outer knob element **360** by the user in a second direction opposite the first direction relieves the compression on the compression element **370** and permits the slider knob **352** to be axially displaced. Accordingly, the design of the actuator **310** permits the user to selectively lock and unlock the axial position of the slider knob **352** along the handle body **332**.

[0070] FIGS. **4**A-**4**C are partial isometric views of a portion of an alternative ablation catheter **400** having a handle and actuator featuring a locking arrangement configured to automatically lock the actuator in one or more discrete, pre-defined axial positions. Except as illustrated and described in connection with FIGS. **4**A-**4**C, the catheter **400** can be constructed in the same manner as the aforementioned catheters **100**, **200** and **300** described above.

[0071] Referring collectively to FIGS. **4**A-**4**C, the handle body of the catheter **400** includes an axial slot **442** bounded by lateral edges, including a lateral edge **444**. The catheter **400** further includes an actuator including a slider knob **452** coupled to a hub **456** that is disposed about and secured to the proximal portion of the inner shaft in the same manner as described above in connection with the ablation catheters **100**, **200** and **300**. Additionally, in the illustrated embodiment, formed in the lateral edge **244** are a plurality of detents **472**, **474** and **476**. The detents **472**, **474** and **476** are spaced from one another along the lateral edge **444** at respective locations corresponding to pre-determined configurations of the spline assembly (see FIGS. **1**B-**1**D). In other embodiments, more or fewer detents may be disposed along the lateral edge **444**.

[0072] In the illustrated embodiment, the aforementioned locking arrangement includes a locking pin **482** having an axial projection **483**, and a torsion spring **484** having a first free end **486** and a second free end **488**. In embodiments, the locking pin **482** is attached to the hub **456** and/or the slider knob **452** and extends outward from the hub **456** through the axial slot **442** to a position adjacent to and in contact with the lateral edge **444**, and the axial projection **483** extends axially from the locking pin **482** inside the interior chamber of the handle.

[0073] Additionally, the torsion spring **484** is disposed about the hub **456**. The first free end **486** slidably engages an inner surface of the handle body, while the second free end **488** is positioned to abut and engage the axial projection **483**. Accordingly, the torsion spring **484** operates to apply a biasing force to the locking pin **482** and thereby bias the locking pin **482** toward the lateral edge **444**, while permitting axial displacement of the slider knob **452**, and thus by extension the inner shaft, when the locking pin **482** is located between any of the detents **472**, **472**, **476**. However, the biasing force applied to the locking pin **482** causes the locking pin **482** to automatically engage the detents **472**, **472**, **476** to releasably lock the axial position of the slider knob **452** at a location corresponding to the respective detent. Additionally, in embodiments, the torsion spring **484** is designed and the axial slot **442** are dimensioned to permit limited rotation of the slider knob **452** sufficient to counteract the biasing force to permit the locking pin **482** to be disengaged from the corresponding detent **472**, **472**, **476**.

[0074] As previously described, the respective locations of the detents **472**, **472**, **476** can be selected so as to automatically and releasably lock the slider knob **452**, and thus the inner shaft, in pre-determined locations corresponding to desired pre-determined configurations of the spline assembly. With reference to FIGS. **1**B-**1**D in addition to FIGS. **4**A-**4**C, for example, in embodiments, the detent **472** may be located to automatically lock the slider knob **452** in the axial position corresponding to the fully expanded configuration of the spline assembly **105** illustrated in FIG. **1**C. Similarly, the detent **474** may be located to automatically lock the slider knob **452** in the axial position corresponding to the partially expanded configuration of the spline assembly **105** illustrated in FIG. **1**D. In the same manner, the detent **476** may be located to automatically lock the slider knob **452** in the axial position corresponding to the fully collapsed configuration of the spline assembly **105** illustrated in FIG. **1**B. Additional configurations of the spline assembly **105** can be accommodated by the inclusion of additional detents.

[0075] In other embodiments, alternative means may be incorporated to automatically or selectively lock the axial position of the actuator mechanism and the inner shaft and, consequently, the configuration of the spline assembly. By way of example only, in one embodiment, the actuator may include a collet mechanism that clamps to the handle body upon rotation of the slider knob to frictionally inhibit axial displacement of the slider knob, and consequently, the inner shaft. In alternative embodiments, the handle body and the slider knob (or other actuator component) may have geometric shapes that create frictional engagement therebetween upon rotation of the slider knob when the user desires to lock the slider knob in place. For example, in one embodiment, the adjacent surfaces of the slider knob and the handle body may each have elliptical profiles, such that rotation of the slider knob relative to the handle body creates an adjustable degree of interference and friction therebetween.

[0076] It is well understood that methods that include one or more steps, the order listed is not a limitation of the claim unless there are explicit or implicit statements to the contrary in the specification or claim itself. It is also well settled that the illustrated methods are just some examples of many examples disclosed, and certain steps may be added or omitted without departing from the scope of this disclosure. Such steps may include incorporating devices, systems, or methods or components thereof as well as what is well understood, routine, and conventional in the art.

[0077] The connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements. The scope is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean "one and only one" unless explicitly so stated, but rather "one or more." Moreover, where a phrase similar to "at least one of A, B, or C" is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B or C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. The terms "couples," "coupled," "connected," "attached," and the like along with variations thereof are used to include both arrangements wherein two or more components are in direct physical contact and arrangements wherein the two or more components are not in direct contact with each other (e.g., the components are "coupled" via at least a third component), but still cooperate or interact with each other.

[0078] In the detailed description herein, references to "one embodiment," "an embodiment," "an example embodiment," etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art with the benefit of the present disclosure to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

[0079] Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present disclosure. For example, while the embodiments described above refer to particular features, the scope of this disclosure also includes embodiments having different combinations of features and embodiments that do not include all of the described features. Accordingly, the scope of the present disclosure is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

We claim:

1. A medical catheter comprising:

a catheter body comprising:

a tubular outer shaft having a proximal end and an open distal end opposite the proximal end;

an inner shaft slidably disposed within the outer shaft having a proximal portion and an opposite distal end, the inner shaft being dimensioned such that the inner shaft distal end extends distally beyond the distal end of the outer shaft;

a spline assembly comprising a plurality of splines each having a spline proximal end attached to the distal end of the outer shaft, and an opposite spline distal end attached to the distal end of the inner shaft, the spline assembly configured to be transitioned between a collapsed configuration dimensioned for slidable insertion through a delivery sheath, and an expanded configuration;

a handle comprising a handle body having a body proximal end, a body distal end, an outer surface extending from the body proximal end to the body distal end, the handle body defining an interior chamber, wherein the proximal end of the outer shaft is attached to and extends from the body proximal end, and the proximal portion of the inner shaft is disposed within the interior chamber; and

an actuator slidably coupled to the handle body between the body proximal end and the body distal end, the actuator comprising a slider knob disposed about and fully circumscribing the outer surface of the handle body such that the slider knob can be axially displaced by a user, wherein the slider knob is further operatively coupled to the proximal portion of the inner shaft such that axial displacement of the slider knob by the user causes the inner shaft to translate axially relative to the outer shaft to transition the spline assembly between the collapsed configuration and the expanded configuration.

2. The medical catheter of claim **1**, wherein the actuator further comprises an inner actuator member attached to and extending from the slider knob into the interior chamber of the handle, and wherein the inner actuator member is fixedly secured to the proximal portion of the inner shaft within the interior chamber and operatively couples the inner shaft to the slider knob.

3. The medical catheter of claim **2** wherein the inner actuator member includes a hub disposed about and fixedly secured to the proximal portion of the inner shaft, and a radial extension extending from the hub to the slider knob.

4. The medical catheter of claim **3**, wherein the handle body includes an axial slot defined by opposed first and second lateral edges, and wherein the radial extension extends through the axial slot and the first and second lateral edges constrain rotation of the inner actuator member during axial displacement of the slider knob.

5. The medical catheter of claim **4**, wherein the actuator is configured such that the user can lock the slider knob in an axial position.

6. The medical catheter of claim **5**, wherein the slider knob includes an outer knob element that is rotatable about the handle body, wherein rotation of the outer knob element by the user in a first direction locks the axial position of the slider knob, and rotation of the outer knob element in a second direction opposite the first direction permits the slider knob to be axially displaced.

7. The medical catheter of claim **6**, wherein the slider knob further comprises an inner knob element that is rotationally fixed relative to the handle body and includes a forward portion disposed radially inward of and in threaded engagement with the outer knob element, wherein rotation of the outer knob element causes the inner knob element to translate axially to selectively lock and unlock the slider knob relative to the handle body.

8. The medical catheter of claim **7**, wherein the outer knob element includes a forward inner surface, and wherein the handle includes a compression element disposed about the handle body between the forward inner surface and the forward portion of the inner knob element, wherein rotation of the outer knob element in the first direction causes the inner knob element to move in a distal direction to compress the compression element between the handle body outer surface, the forward inner surface of the outer knob element, and the forward portion of the inner knob element to lock the axial position of the slider knob.

9. The medical catheter of claim **4**, wherein the first lateral edge includes a first detent at a first location along the lateral edge, and wherein the locking arrangement comprises a locking pin extending radially from the hub through the axial slot, wherein the locking pin is biased toward the first lateral edge so as to automatically engage the first detent to releasably lock the axial position of the slider knob corresponding to the first location.

10. The medical catheter of claim **9**, wherein the locking arrangement further includes a torsion spring disposed about the hub, the torsion spring including a first free end frictionally engaged with an inner surface of the handle body, and an opposite second free end positioned to engage and apply a biasing force to the locking pin to bias the locking pin toward the first lateral edge.

11. The medical catheter of claim **10**, wherein the slider knob is configured to be rotatable to counteract the biasing force and selectively disengage the locking pin from the first detent.

12. The medical catheter of claim **11**, wherein the first lateral edge includes a second detent at a second location along the first lateral edge, and a third detent at a third location along the first lateral edge, the first, second and third locations being spaced from one another and corresponding to pre-determined configurations of the spline assembly.

13. A medical catheter comprising:

a catheter body comprising a tubular outer shaft and an inner shaft slidably disposed within the outer shaft and dimensioned to extend distally beyond the outer shaft;

a radially expandable and radially collapsible spline assembly comprising a plurality of splines each having a spline proximal end attached to the distal end of the outer shaft, and an opposite spline distal end attached to the distal end of the inner shaft;

a handle coupled to the catheter body, the handle having a handle proximal end and a handle distal end, wherein a proximal end of the outer shaft is attached to and extends from the handle proximal end, and a proximal portion of the inner shaft is slidably disposed within the handle; and

an actuator comprising a slider knob slidably disposed about and fully circumscribing an outer surface of the handle, wherein the slider knob is operatively coupled to the proximal portion of the inner shaft such that axial displacement of the slider knob by a user causes the inner shaft to translate axially relative to the outer shaft to selectively control an amount of expansion of the spline assembly.

14. The medical catheter of claim **13**, wherein the actuator is configured such that the user can lock the slider knob in an axial position.

15. The medical catheter of claim **13**, wherein the actuator further comprises a locking arrangement configured to releasably lock the actuator in one or more pre-defined axial positions.

16. A handle assembly for a medical catheter having a catheter body including a tubular outer shaft, an inner shaft slidably disposed within the outer shaft, and a radially expandable and collapsible splined electrode assembly disposed at a distal end of the catheter body, the handle assembly comprising:

a handle comprising a handle body having a body proximal end, a body distal end, an outer surface extending from the body proximal end to the body distal end, the handle body defining an interior chamber configured to slidably receive a proximal portion of the inner shaft; and

an actuator slidably coupled to the handle body between the body proximal end and the body distal end, the actuator comprising a slider knob disposed about and fully circumscribing the outer surface of the handle body such that the slider knob can be axially displaced by a user, wherein the slider knob is further configured to be operatively coupled to the proximal portion of the inner shaft such that axial displacement of the slider knob by the user can cause the inner shaft to translate axially relative to the outer shaft to transition the spline assembly between a collapsed configuration and an expanded configuration.

17. The handle assembly of claim **16**, wherein the handle body includes an axial slot defined by opposed first and second lateral edges, and the actuator further comprises an inner actuator member including a hub configured to be disposed about and fixedly secured to the proximal portion of the inner shaft, and a radial extension extending from the hub through the axial slot to the slider knob.

18. The handle assembly of claim **17**, wherein the slider knob includes an outer knob element that is rotatable about the handle body, an inner knob element that is rotationally fixed relative to the handle body and includes a forward portion disposed radially inward of and in threaded engagement with the outer knob element, wherein rotation of the outer knob element causes the inner knob element to translate axially to selectively lock and unlock the slider knob relative to the handle body.

19. The handle assembly of claim **18**, wherein the outer knob element includes a forward inner surface, and wherein the handle further includes a compression element disposed about the handle body between the forward inner surface and the forward portion of the inner knob element, wherein rotation of the outer knob element in a first direction causes the inner knob element to move in a distal direction to compress the compression element between the handle body outer surface, the forward inner surface of the outer knob element, and the forward portion of the inner knob element to lock the axial position of the slider knob.

20. The handle assembly of claim **17**, wherein the first lateral edge includes a plurality of axially-spaced detents, and wherein the actuator includes a locking arrangement comprises a locking pin extending radially from the hub through the axial slot, wherein the locking pin is biased toward the first lateral edge so as to automatically engage the one of the plurality of detents to releasably lock the axial position of the slider knob corresponding to a predetermined configuration of the spline assembly.

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